RiskPulse

From headlines to stress tests: an AI/NLP risk engine with event-driven portfolio stress testing

Case study: Module B, Strategic Portfolio Stress Testing

Problem and approach

Risk teams read the news; the portfolio only moves when someone runs the numbers

1 Read

Ingest live news headlines and social posts and normalise them into one document schema.

2 Score

The NLP engine returns a sentiment score, an event class and an impact score from 1 to 10 for every item.

3 Stress

Any signal with impact above 7 triggers a stress test on a synthetic wholesale-banking book.

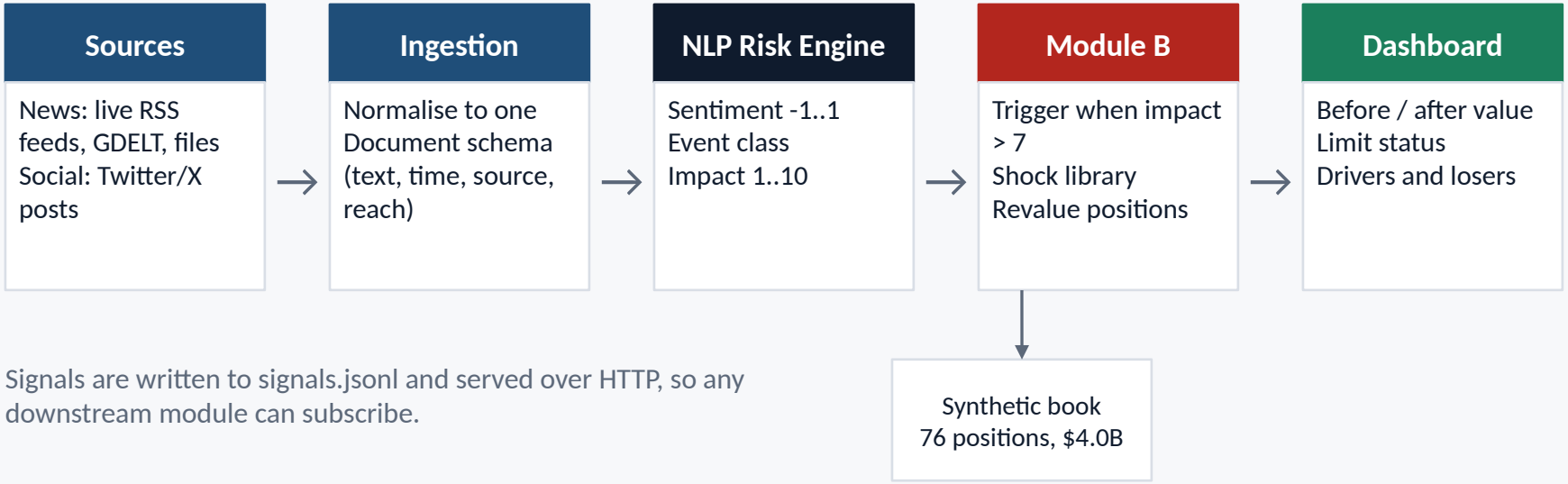
4 Show

A dashboard compares portfolio value before and after, with loss against risk limits.

Why it matters: the gap between a headline and a quantified, limit-checked loss estimate shrinks from hours to seconds, and every number traces back to the text that caused it.

System design

One pipeline from raw text to a limit-checked loss estimate



Signals are written to signals.jsonl and served over HTTP, so any downstream module can subscribe.

The Risk Engine

Transparent scoring: every number can be explained to a risk reviewer

Sentiment Finance lexicon with negation and intensifiers, squashed to -1..1. FinBERT optional.

Event class Weighted patterns over 10 classes, with a confidence. Geopolitical, Credit, Macro, Cyber, M&A and more.

Impact 1-10 Event base severity + sentiment (negative weighs more) + severity terms + amounts and % + social reach, scaled by source credibility.

Input (news)

Regional lender defaults on $4 billion of bonds; creditors seek restructuring.

Output (signal)

```
event_type: Credit Event
sentiment_score: -0.96
impact_score: 10.0
entities: [DB]  region: Europe
rationale: base 6.5, sentiment +2.4,
severity +1.2, size +0.5
```

Module B: stress testing

Event class picks the shock set; impact score sets how hard it hits

Event (full severity)	Equity	Yields	Spreads	Notes
Geopolitical	-15%	-50 bp	+110 bp	EM FX -7%, JPY and CHF up
Macroeconomic	-10%	+125 bp	+70 bp	Real estate and tech hit harder
Credit Event	-12%	-25 bp	+180 bp	Financials 1.7x
Cyber/Operational	-8%	-10 bp	+60 bp	Financials and tech
Natural Disaster	-7%	-5 bp	+80 bp	2x in region, 0.4x elsewhere

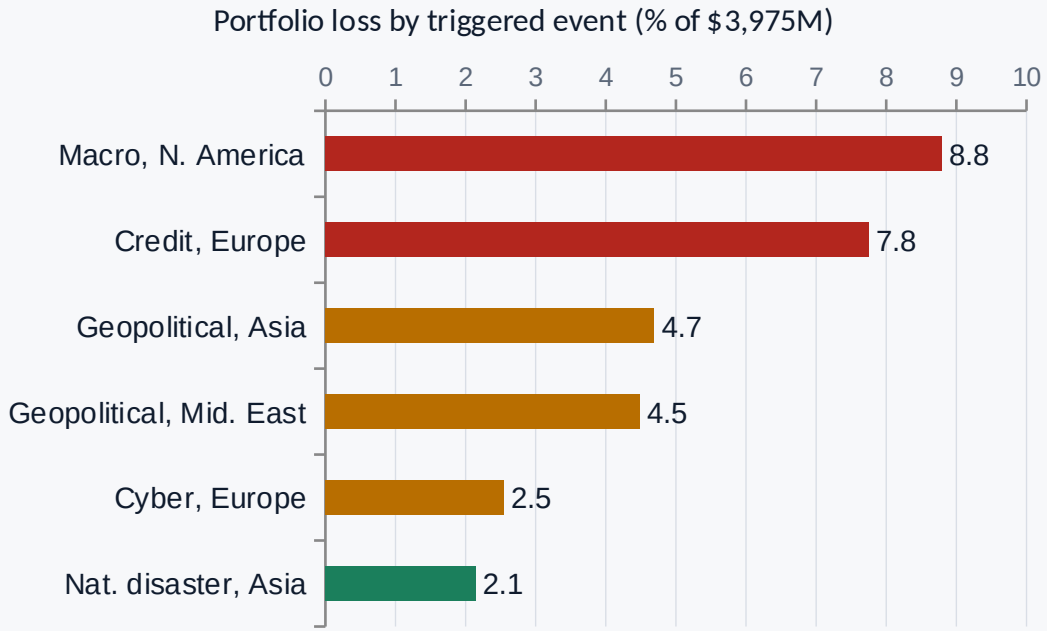
Revaluation by asset type

- Bonds: duration-convexity plus spread duration
- Loans: spread mark-down plus PD-scaled expected loss
- Swaps (DV01) and CVA, FX forwards, options (delta-gamma)
- P&L split into rates, credit, equity and FX

Limits: WARNING beyond -2.5% of portfolio value, BREACH beyond -5%.

Results on the sample feed

6 of 23 signals crossed the threshold; 2 breached the 5% loss limit



What the run shows

- Worst case: a macro rate shock costs $349M (8.8%)
- Credit contagion costs $308M, mostly in bonds
- News and Twitter corroborate each trigger, so one event is not run twice
- Red = breach, amber = warning, green = within limits

Domain impact and next steps

Impact

- Turns unstructured news into limit-checked P&L estimates in seconds
- Explainable end to end: signal rationale, shock table, risk-factor attribution
- Modular: the engine's signal format also feeds Module A rebalancing

Limitations and next steps

- Rule-based NLP misses sarcasm: fine-tune on the Kaggle financial news and tweet datasets
- Shocks are assumptions: calibrate to realised moves after historical GDELT events
- First-order pricing and additive combined loss: add correlations and full revaluation

Prototype only: not a regulatory capital model. All data is public or synthetic.